

# Socioeconomic impacts of Terrestrial Protected Areas

## Evidence from Large-Scale National Surveys in Madagascar

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# To what extent do protected areas impact household living standards?

▷ Evolution of protected areas (graphics)

## ▶ **Economic Opportunities**

- ▶ Development of local tourism (Naidoo et al. 2019),
- ▶ Creation of conservation-related jobs (Balietti et al. 2024)
- ▶ Compensation measures (Poudyal et al. 2018)
- ▶ Improved ecosystem services (Xu et al. 2023)

## ▶ **Restrictions and Costs** Restricted access to natural resources - gathering, hunting, fishing, medicinal plants (Kandel et al. 2022)

## ▶ **Heterogeneous results** Effects vary across evaluation methods, institutional contexts, and geographic locations (Kandel et al. 2022).

# Research design

## Hypothesis

- Protected Areas (PAs) reduce average household wealth?
- Protected Areas (PAs) increase socioeconomic inequalities within communities by differentially channeling benefits towards better positioned households?
- Multi-use PAs (IUCN categories V-VI) generate more favorable outcomes than strict PAs (IUCN I-IV)?

## Research Question

To what extent do protected areas affect household living standards?

## Contribution

- National analysis of 84 protected areas.
- Quantitative assessment of the impact of protected area creation on household living conditions.

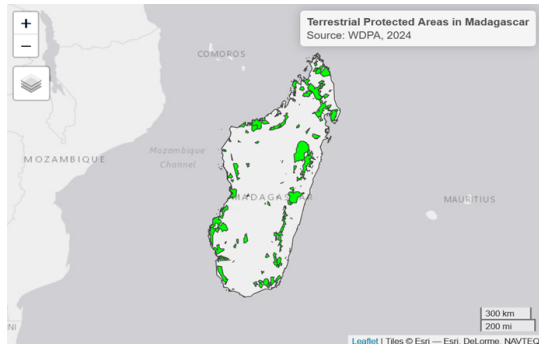
# Madagascar is a relevant study case

## Area of protected areas and Size of the adjacent population exposed to them nearly tripled between 2008 and 2021

- ▶ 2008: 3,6% of the territory — 9% population lived < 10 km of PAs
- ▶ 2021: 10,8% du territoire — 28% population lived > 10 km of PAs

## A Vulnerable socio-economic context

- ▶ The world's poorest country based on SDG Indicator 1.1 (Conceição 2024)
- ▶ High dependence on natural resources (Llopis et al. 2023)
- ▶ Weak state capacity (Hanson and Sigman 2021)



Source : Authors

# Theory of Change

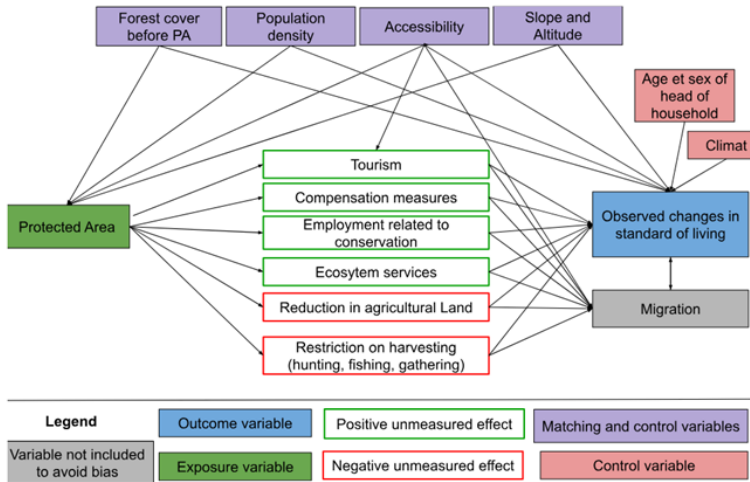


Figure 1: Logic diagram of the theory of change

Empirical methodology

## The data at the rural level

- ▶ **Dependent variable:** Protected Areas Implementation, from WDPA
- ▶ **Outcome variable:** Wealth index and z-score of the wealth index from Demographic and Health surveys (DHS) (1997, 2008, 2021), Malaria Indicator Surveys (MIS) (2011, 2013, 2016), and Multiple Indicator Cluster Surveys (MICS) (2018)
- ▶ **Matching variables:** forest cover rates in 2000 (m), slope, elevation (%), population density (km<sup>2</sup>), accessibility (min), and standardized precipitation evapotranspiration index, downloaded via [mapme.biodiversity](https://mapme.biodiversity.org/)
- ▶ **Control variables:** age and sex of the head of household, from DHS, MIS, MICS
- ▶ **Treatment group:** households living in rural areas < 10 km of a Protected areas created between 2008 and 2021
- ▶ **Control group:** households living in rural areas < 10 km of a Protected areas created between 2008 and 2021

# Baseline model: protected areas, living conditions, and rural

## ► Genetic matching

Used to compare observable characteristics and increase the similarity between the treatment group and the control group (Diamond and Sekhon 2013)

## ► Difference-in-difference (DID)

### i Baseline Equation

L'équation de la méthode de difference-in-difference (Daw and Hatfield 2018):

$$Y_{ict} = \alpha + \delta \cdot (\text{Treatment}_c \times \text{Post}_t) + X'_{ict}\theta + \mu_c + \lambda_t + \varepsilon_{ict}$$

►  $Y_{ict}$ : wealth index (household  $i$ , cluster  $c$ , year  $t$  1997/2008/2021)

►  $\alpha$ : intercept (mean control group pre-treatment)

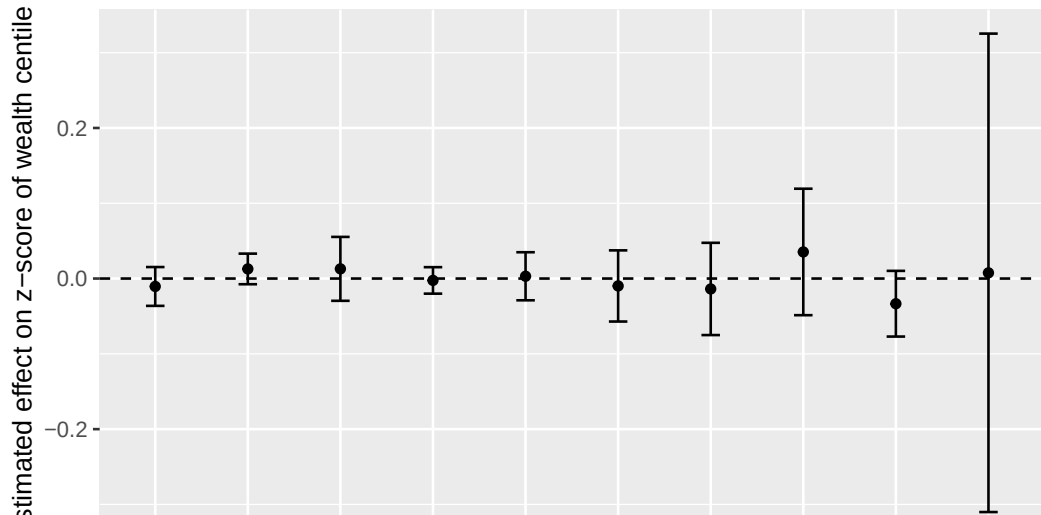
►  $\text{Treatment}_c = 1$  if cluster affected by protected area, 0 otherwise



## Results

## Protected Areas impact on the household wealth index

# Protected Areas impact on the household wealth index inequalities



# Heterogeneity: Which Protected Areas Harm Most?

## Robustness Test

- ▶ **Distance threshold test of 5 km and 15km:** Proximity to protected areas (PAs) has no significant effect on wealth
- ▶ **Pseudo-panel analysis** the results confirm the robustness of the conclusions in that environmental and spatial conditions have a stronger influence on household living standards than sociodemographic characteristics
- ▶ **Heterogeneity analysis:** The effects are not robust to the presence of unobserved bias
- ▶ **Other outcome variables (Child size/weight and infant mortality):** No significant effect is found in the estimation

## Conclusions

- ▶ PAs do not lower the standard of living, but costs accumulate over time
- ▶ No significant average effect on the standards of living - losses associated with restrictions appear to be offset by adaptation mechanisms and local opportunities
- ▶ No significant effect on inequality - the main determinants remain climate shocks and sociodemographic characteristics
- ▶ No differentiated effect based on governance type - formal categories alone do not explain socioeconomic outcomes

**\*\*Policy Implications\*** - Integration of systematic socioeconomic monitoring into protected area governance frameworks - Strengthening of transparent compensation mechanisms - Design of inclusive benefit-sharing mechanisms that address equity concerns

## References

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<https://doi.org/10.1016/j.jeem.2024.103070>.
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- Diamond, Alexis, and Jasjeet S. Sekhon. 2013. "Genetic Matching for Estimating Causal Effects: A General Multivariate Matching Method for Achieving Balance in

# Appendix



# Evolution of protected areas (graphics)

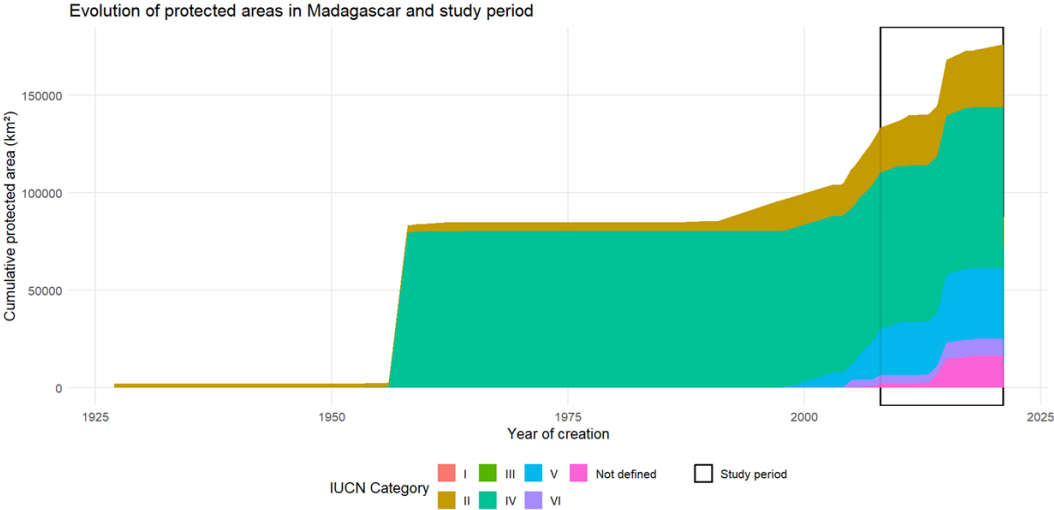


Figure 2. Evolution of protected areas in Madagascar and study period